

continuous $\bar{g}^{-1}(U) \in \mathcal{V}$.

Since $f: (X, \mathcal{T}) \rightarrow (Y, \mathcal{V})$ is $\mathcal{T}-\mathcal{V}$

continuous $\bar{f}^{-1}(\bar{g}^{-1}(U)) = (g \circ f)^{-1}(U) \in \mathcal{T}$.

Theorem: Let $(X, \mathcal{T})$ and $(Y, \mathcal{V})$ be two topological spaces & $f: X \rightarrow Y$ be a $\mathcal{T}-\mathcal{V}$ continuous function. Let $(X, \mathcal{T}_1)$ & $(Y, \mathcal{V}_1)$ be the topological spaces s.t. $\mathcal{T} \subseteq \mathcal{T}_1$ and $\mathcal{V}_1 \subseteq \mathcal{V}$. Then $f: X \rightarrow Y$ is $\mathcal{T}_1-\mathcal{V}_1$ continuous.

Proof:

Let $V \in \mathcal{V}_1$. Since $\mathcal{V}_1 \subseteq \mathcal{V}$, $V \in \mathcal{V}$.

As $f: X \rightarrow Y$ is $\mathcal{T}-\mathcal{V}$ continuous

$f^{-1}(V) \in \mathcal{T}$. Because $\mathcal{T} \subseteq \mathcal{T}_1$

$\bar{f}^{-1}(V) \in \mathcal{T}_1$.

$\therefore f$ is $\mathcal{T}_1-\mathcal{V}_1$ continuous.

Theorem: Let $(X, \mathcal{C})$, $(Y, \mathcal{V})$ be topological spaces & $f: X \rightarrow Y$ be a function. Then f is continuous if and only if the inverse image $f^{-1}(K)$ of every $\mathcal{V}$ -closed subset of Y under f is a $\mathcal{C}$ -closed subset of X .

Proof:

$\Rightarrow$ Let $K \subseteq Y$ be a $\mathcal{V}$ -closed subset of Y . Thus $Y \setminus K \in \mathcal{V}$. Since f is $\mathcal{C}$ - $\mathcal{V}$ continuous $f^{-1}(Y \setminus K) \in \mathcal{C}$. But $f^{-1}(Y \setminus K) = X \setminus f^{-1}(K) \in \mathcal{C}$. Hence $f^{-1}(K)$ is a $\mathcal{C}$ -closed subset of X .

$\Leftarrow$ Let V be a $\mathcal{V}$ -open subset of Y .

Then $Y \setminus V \in \mathcal{V}_K$. By hypothesis

$$f^{-1}(Y \setminus V) = X \setminus f^{-1}(V) \in \mathcal{C}_K.$$

$$\Rightarrow f^{-1}(V) \in \mathcal{C}.$$

$\therefore f: X \rightarrow Y$ is a $\mathcal{C}$ - $\mathcal{V}$

continuous function.

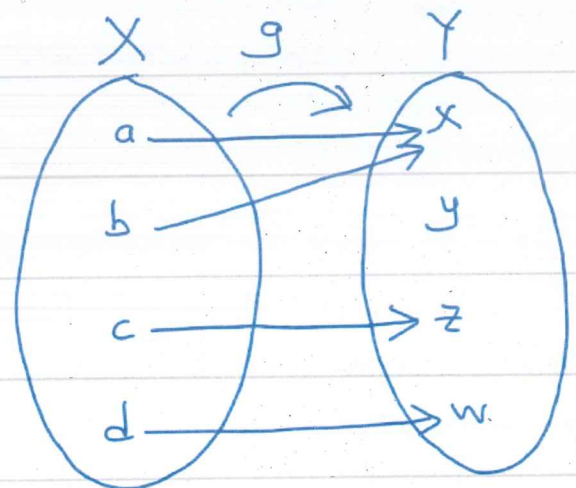
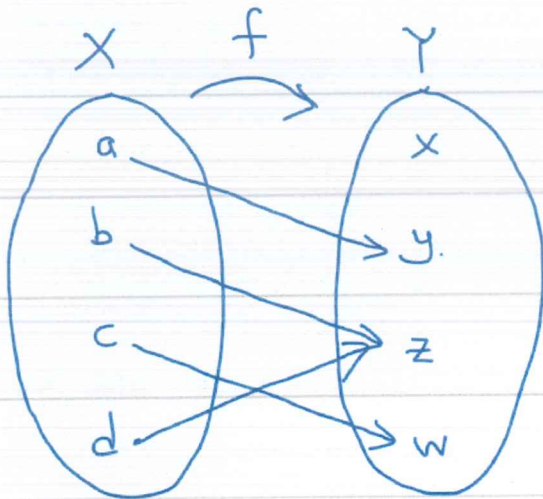
Examples;

1. Consider the following topologies on $X = \{a, b, c, d\}$ & $Y = \{x, y, z, w\}$, respectively:

$$\mathcal{T} = \{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}, X\}$$

$$\mathcal{V} = \{\emptyset, \{x\}, \{y\}, \{x, y\}, \{y, z, w\}, Y\}.$$

Moreover consider the functions $f: X \rightarrow Y$ and $g: X \rightarrow Y$ defined by the diagrams below:



Determine if the functions f & g are $\mathcal{T}$ - $\mathcal{V}$ continuous.

Solution;

$$f^{-1}(\emptyset) = \emptyset \in \mathcal{T}$$

$$f^{-1}(\{x\}) = \emptyset \in \mathcal{T}$$

$$f^{-1}(\{y\}) = \{a\} \in \mathcal{T}$$

$$f^{-1}(\{x, y\}) = \{a\} \in \mathcal{T}$$

$$f^{-1}(\{y, z, w\}) = X \in \mathcal{T}$$

$$f^{-1}(Y) = X \in \mathcal{T}.$$

$\therefore f$ is τ - $\mathcal{V}$ continuous.

Because for $\{y, z, w\} \in \mathcal{V}$,
 $g^{-1}(\{y, z, w\}) = \{c, d\} \notin \tau$.

$\therefore g$ is not τ - $\mathcal{V}$ continuous.

2. Let (X, τ) be a discrete topological space & $(Y, \mathcal{V})$ be an arbitrary topological space. Then every function $f: X \rightarrow Y$ is τ - $\mathcal{V}$ continuous.

3. Let (X, τ) and $(Y, \mathcal{V})$ be two topological spaces. If $f: X \rightarrow Y$ is a constant function then show that f is τ - $\mathcal{V}$ continuous.

Solution:

Since f is a constant function there exists a $y_0 \in Y$ s.t. $f(x) = y_0$ for all $x \in X$.

Let $V \in \mathcal{V}$.

Case 1 If $y_0 \in V$ then
 $\bar{f}^{-1}(V) = X \in \mathcal{T}$.

Case 2 Assume that $y_0 \notin V$. Then
 $\bar{f}^{-1}(V) = \emptyset \in \mathcal{T}$.

$\therefore f$ is $\mathcal{T}$ - $\mathcal{V}$ continuous.

4. Let (X, τ) be any topological space and (Y, ϑ) be an indiscrete topological space. Then any function $f: X \rightarrow Y$ is τ - ϑ continuous.

5. Let (X, τ) and (Y, ϑ) be cofinite topological spaces and $f: X \rightarrow Y$ be a nonconstant function. Show that $f: X \rightarrow Y$ is τ - ϑ continuous $\Leftrightarrow f^{-1}(\{y\})$ is finite for every $y \in Y$.

Solution:

Recall that a set in a cofinite topological space is closed iff that set is finite.

$\Rightarrow$ Let $y \in Y$. Because $\{y\}$ is finite $\{y\}$ is a ϑ -closed subset of Y . Since f is τ - ϑ continuous $f^{-1}(\{y\}) \in \tau$.

Hence $f^{-1}(\{y\})$ is finite. ($f^{-1}(\{y\})$ is either

X or it is finite. But f is nonconstant. Hence $f^{-1}(\{y\}) \neq X$)

$\Leftarrow$ Let $K \subseteq Y$ be a $\mathcal{V}$ -closed subset of Y . Since $\mathcal{V}$ is the cofinite topology on Y , K is closed iff K is finite.

$$\text{Assume that } K = \{y_1, y_2, \dots, y_n\} \\ = \bigcup_{i=1}^n \{y_i\}.$$

Then

$$f^{-1}(K) = f^{-1}\left(\bigcup_{i=1}^n \{y_i\}\right) = \bigcup_{i=1}^n \underbrace{f^{-1}(\{y_i\})}_{\text{is finite}} \text{ is finite.}$$

is finite

by our hypothesis

$$\Rightarrow f^{-1}(K) \text{ is } \mathcal{T}\text{-closed.}$$

$$\Rightarrow f \text{ is } \mathcal{T}\text{-}\mathcal{V} \text{ continuous.}$$

$$\left(\text{If } K = Y \in \mathcal{V}_K \text{ then } f^{-1}(K) = X \in \mathcal{T}_K \right).$$

6. Let $(X, \mathcal{T}_1)$ & $(X, \mathcal{T}_2)$ be topological spaces. Consider the function $i : X \rightarrow X$. Then $x \mapsto i(x) = x$.

$$i \text{ is } \mathcal{T}_1\text{-}\mathcal{T}_2 \text{ continuous} \Leftrightarrow \mathcal{T}_2 \subseteq \mathcal{T}_1.$$

(WHY?)

Definition: Let (X, τ) and (Y, ϑ) be topological spaces & $f: X \rightarrow Y$ be a function. f is called an open (respectively closed) function if the image of every τ -open (resp. τ -closed) set is ϑ -open (resp. ϑ -closed).

In general functions which are open need not be closed and vice versa.

Also in general, functions which are open (resp. closed) need not be continuous. Similarly a continuous function need not be open (resp. closed).

Note;

Look at Example 1 given above.

In that example f is a continuous function. But if you check you will see that f is not an open function.

Example: Let $X = \{a, b, c, d\}$,

$Y = \{x, y, z\}$ be sets,

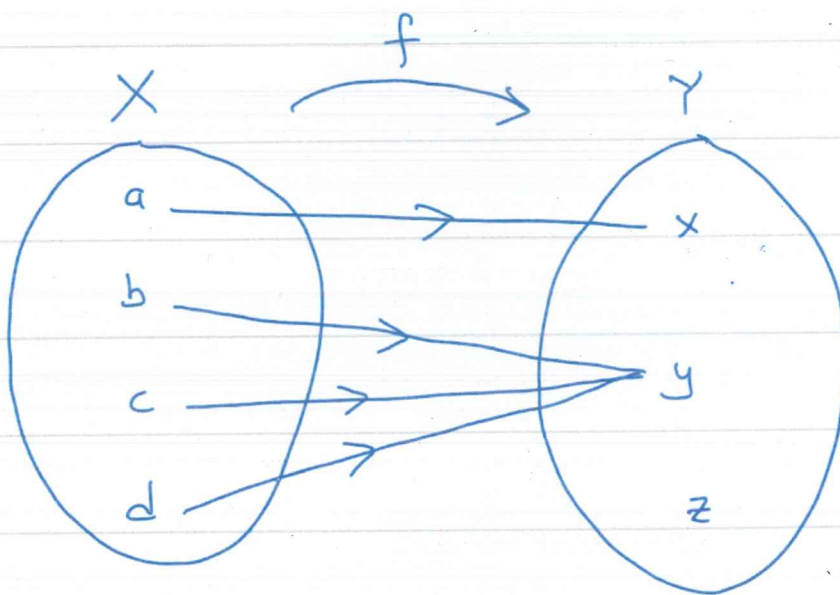
$\tau = \{\emptyset, \{a\}, \{b\}, \{d\}, \{a, b\}, \{a, d\}, \{b, d\}, \{a, b, d\}, X\}$ and

$\mathcal{V} = \{\emptyset, \{x\}, \{y\}, \{x, y\}, Y\}$

be topologies on X & Y , respectively.

Consider the function $f: X \rightarrow Y$ given

by



(i) Is f an open function?

(ii) Is f a closed function?

(iii) Is f a continuous function?

Solution:

$$(i) f(\emptyset) = \emptyset \in \mathcal{V} \quad f(\{a, b, d\}) = \{x, y\} \in \mathcal{V}$$

$$f(\{a\}) = \{x\} \in \mathcal{V} \quad f(X) = \{x, y\} \in \mathcal{V}$$

$$f(\{b\}) = \{y\} \in \mathcal{V}$$

$$f(\{d\}) = \{y\} \in \mathcal{V}$$

$$f(\{a, b\}) = \{x, y\} \in \mathcal{V}$$

$$f(\{a, d\}) = \{x, y\} \in \mathcal{V}$$

$$f(\{b, d\}) = \{y\} \in \mathcal{V}$$

$\therefore f$ is an open function.

$$(ii) \mathcal{C}_X = \{X, \{b, c, d\}, \{a, c, d\}, \{a, b, c\}, \{c, d\}, \{b, c\}, \{a, c\}, \{c\}, \emptyset\}$$

$$\mathcal{V}_K = \{Y, \{y, z\}, \{x, z\}, \{z\}, \emptyset\}$$

$$f(X) = \{x, y\} \notin \mathcal{V}_K$$

$$f(\{b, c, d\}) = \{y\}$$

$$f(\{a, c, d\}) = \{x, y\}$$

$$f(\{a, b, c\}) = \{x, y\}$$

$$f(\{c, d\}) = \{y\}$$

$$f(\{b, c\}) = \{y\}$$

$$f(\{a, c\}) = \{x, y\}$$

$$f(\{c\}) = \{y\}$$

$$f(\emptyset) = \emptyset$$

Since $X \in \mathcal{T}_K$ but $f(X) \notin \mathcal{V}_K$

f is not a closed function.

(iii) $\{y\} \in \mathcal{V}$. But $f^{-1}(\{y\}) = \{b, c, d\} \notin \mathcal{T}$.

$\therefore f$ is discontinuous.

Definition: Two topological spaces $(X, \mathcal{T})$ & $(Y, \mathcal{V})$ are called

homeomorphic or topologically equivalent

if there exists a bijection $f: X \rightarrow Y$

s.t. both f & f^{-1} are continuous. Such

a function f is called a homeomorphism.

If $(X, \mathcal{T})$ & $(Y, \mathcal{V})$ are homeomorphic

then we write $X \cong Y$.